

A Dynamical–Spectral Theory of Knowledge Organization in Large Language Model Parameter Spaces

Toward a Mathematical Foundation for Knowledge–Parameter Topology and Continual
Parameter Plasticity

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摘要

Large Language Models (LLMs) are conventionally described by a fixed parameter vector θ obtained after large-scale optimization. Such a description is useful for training, but it provides an incomplete theoretical picture of continual learning: when a model encounters new knowledge, the relevant internal parameter state may change, yet there is no generally accepted mathematical framework for describing how knowledge organization, parameter structure, and parameter dynamics are related.

This paper proposes a dynamical–spectral theoretical framework in which an LLM parameter system is treated as a high-dimensional, nonlinear dynamical system rather than merely a static collection of optimization variables. The framework introduces three coupled spaces: a semantic knowledge space $\mathcal{K}$, a parameter space $\mathcal{P}$, and a spectral state space $\mathcal{S}$. The central object is a time-dependent parameter state $\theta(t)$, whose evolution is represented by $\dot{\theta} = F(\theta, x, r, u)$. The local dynamics are characterized by a Jacobian $J_\theta = \partial F / \partial \theta$, while the spectral state is obtained from the eigenvalue, singular-value, eigenvector, and invariant-subspace structure of parameter matrices and their Jacobians.

The theory does not assert that a single eigenvalue directly “means” a particular fact. Instead, it proposes a stronger and more testable statement: if knowledge is encoded in the model’s parameters, then changes in knowledge necessarily induce changes in the parameter state, and these changes can be studied through mathematically defined spectral observables. Knowledge regions are therefore modeled as distributed parameter subspaces with potentially overlapping boundaries, while knowledge evolution is modeled as a trajectory on a parameter–spectral manifold. The framework distinguishes existence of a mapping from its identifiability and stability, and introduces observability, perturbation, subspace-angle, Fisher-information, Hessian, and Jacobian-spectrum tools for empirical validation.

This theory provides a mathematical bridge between two prior conceptual works, Reinforcement-Learned Knowledge–Parameter Topology (RL-KPT), which hypothesizes structured knowledge-to-parameter regions and localized parameter plasticity, and RL-KPT-SI, which investigates

the mathematical feasibility of Jacobian-based spectral analysis of LLM parameter systems. In the present framework, RL is not treated as the source of the dynamical theory; rather, it can later act as an active controller whose state observation is supplied by the spectral-dynamical analysis developed here. The resulting hierarchy is

knowledge space $\longrightarrow$ parameter state $\longrightarrow$ Jacobian/spectral state $\longrightarrow$ knowledge dynamics, with reinforcement learning optionally providing the active control layer for continual adaptation.

The paper formulates definitions, propositions, measurable hypotheses, a practical experimental protocol, and limitations. It is intended as a theoretical foundation rather than a claim that the proposed knowledge-spectral correspondence has already been empirically established.

Keywords: Large Language Models; parameter dynamics; knowledge-parameter topology; spectral analysis; Jacobian matrix; continual learning; parameter plasticity; dynamical systems; reinforcement learning; knowledge localization.

1 Introduction

Large Language Models have demonstrated that a very large neural parameter system can acquire linguistic, factual, procedural, mathematical, and domain-specific capabilities through large-scale optimization. The standard mathematical abstraction is

$$y = f_{\theta}(x),$$

where x is an input, y is a model output, and θ is the learned parameter vector.

After pretraining, however, θ is usually treated as fixed:

$$\theta_{t+1} \approx \theta_t.$$

New information can be provided through context, retrieval, external memory, fine-tuning, model editing, or parameter-efficient adaptation, but these mechanisms do not by themselves provide a general theory of how the internal parameter system reorganizes when knowledge changes.

This motivates a more fundamental question:

Can the internal parameter system of an LLM be modeled as a structured dynamical system in which knowledge organization and knowledge evolution are represented by measurable properties of parameter and spectral states?

The present paper develops a theoretical answer in the form of a research framework.

The starting point is the hypothesis developed in RL-KPT: a sufficiently trained language model may contain structured relationships between semantic knowledge states and functional regions of its parameter space. RL-KPT therefore defines a semantic knowledge coordinate system and hypothesizes a mapping

$$F : \mathcal{K} \times \mathcal{H} \rightarrow \mathcal{P},$$

where $\mathcal{K}$ denotes semantic knowledge states, $\mathcal{H}$ denotes internal model states, and $\mathcal{P}$ denotes controllable parameter regions. The framework explicitly allows distributed and overlapping regions rather than requiring a one-to-one mapping between facts and individual parameters. In particular, interdisciplinary knowledge may correspond to overlapping subspaces such as

$$P_{\text{robotics}} \approx P_{\text{mechanics}} \cup P_{\text{control}} \cup P_{\text{physics}} \cup P_{\text{mathematics}}.$$

RL-KPT-SI subsequently investigates the use of Jacobian matrices as a mathematically tractable route to spectral analysis of LLM parameter systems. The present paper supplies the missing theoretical layer: it treats the parameter system itself as a dynamical object and asks how its Jacobian and spectral state evolve when knowledge changes.

The main thesis is therefore not that “an eigenvalue equals a knowledge fact.” Rather, the thesis is that knowledge, parameter state, and spectral state form a coupled system:

$$K(t) = \Phi(\theta(t)), \quad S(t) = \Psi(\theta(t), J_\theta(t)),$$

and consequently

$$\dot{K}(t) = D\Phi(\theta(t))\dot{\theta}(t), \quad \dot{S}(t) = D\Psi(\theta(t), J_\theta(t), \dot{J}_\theta(t)).$$

The central scientific object becomes the *dynamic mapping*

$$\boxed{K(t) \longleftrightarrow \theta(t) \longleftrightarrow S(t)}$$

rather than a static lookup table between a fact and a parameter.

This shift is important for continual learning. If knowledge changes continuously, then the relevant mathematical problem is not merely to locate a static parameter region, but to measure which parameter subspaces move, which spectral quantities change, which regions remain stable, and which regions are newly recruited.

2 Relation to Previous Work

2.1 RL-KPT: Knowledge–Parameter Topology

RL-KPT proposes that semantic knowledge can be represented in an explicit knowledge space

$$\mathcal{K} = \{K_1, K_2, \dots, K_m\},$$

while the parameter system can be represented by controllable functional regions

$$\mathcal{P} = \{P_1, P_2, \dots, P_q\}.$$

A knowledge item x is encoded as a semantic vector

$$z_x = f_K(x).$$

The proposed routing policy has the form

$$M_t = \pi_\phi(S_t, K_t),$$

and localized adaptation is written as

$$\theta_{t+1} = \theta_t + M_t \odot \Delta\theta_t.$$

The essential conceptual change is from asking only how to optimize parameters to asking *where* new knowledge should be incorporated.

RL-KPT deliberately leaves open whether the hypothesized topology exists, how localized it is, and how stable it remains when knowledge accumulates. Those open questions motivate the present theory.

2.2 RL-KPT-SI: Jacobian and Spectral Analysis

RL-KPT-SI considers the mathematical feasibility of treating Jacobian matrices as objects for spectral analysis of LLM parameter systems. This introduces a structural analysis route of the form

$$\theta \longrightarrow J(\theta) \longrightarrow \sigma(J),$$

where $\sigma(J)$ denotes a collection of spectral observables.

The present paper extends this idea from a static mathematical operation to a dynamical framework:

$$\theta(t) \longrightarrow J(t) \longrightarrow \sigma(J(t)).$$

The object of interest is consequently not merely the spectrum at one point, but its trajectory, perturbation, and relationship to knowledge-state changes.

2.3 Position Relative to Existing Neural Dynamics

Neural ODEs model hidden-state evolution through differential equations and provide a mathematical language for continuous-depth neural computation [?]. That literature establishes the usefulness of dynamical-systems descriptions of neural networks, but the present paper focuses on a different state variable: the *parameter system itself*. The proposed state is therefore

$$\theta(t) \in \mathcal{T}$$

rather than only a hidden activation trajectory.

Spectral analysis of neural-network weight matrices is also established. Random-matrix studies have shown that singular-value statistics can distinguish structured large singular directions from approximately random components in trained networks [?]. The present framework uses such spectral information as a potential state representation for knowledge organization and evolution rather than treating spectrum solely as a measure of network complexity.

Mechanistic interpretability and knowledge-localization research has demonstrated that specific concepts, features, neurons, layers, or directions can sometimes be localized or probed in

language models. However, localization does not automatically imply necessity, uniqueness, or causal sufficiency. This motivates a dynamical formulation in which localization is tested through perturbation, subspace structure, and temporal evolution rather than by static correlation alone.

3 Mathematical Representation of an LLM Parameter System

3.1 Parameter State

Let an LLM consist of L layers with parameter matrices

$$W_l \in \mathbb{R}^{d_l \times d'_l}, \quad l = 1, \dots, L.$$

The complete parameter state is

$$\theta = \text{vec}(W_1, \dots, W_L) \in \mathbb{R}^N.$$

The parameter space is therefore

$$\mathcal{T} \subseteq \mathbb{R}^N.$$

The use of θ as a state variable is a conceptual change from the conventional optimization-only view. The question is no longer merely whether θ minimizes a loss, but how θ evolves as the model encounters experience.

3.2 Knowledge State

Let

$$\mathcal{K} \subseteq \mathbb{R}^m$$

be a semantic knowledge space whose coordinates correspond to human-interpretable high-level domains or learned semantic axes.

A knowledge item x is mapped to

$$z = f_K(x) \in \mathcal{K}.$$

The coordinate system can be hierarchical. For example,

$$K_{\text{engineering}} \supset \{K_{\text{mechanics}}, K_{\text{control}}, K_{\text{electrical}}, \dots\}.$$

The theory does not require the hierarchy to be complete or universal; it only requires that the chosen semantic coordinate system be explicit and reproducible.

3.3 Parameter Regions

A parameter region is a subset

$$P_i \subseteq \mathcal{T}.$$

Regions may be defined at multiple scales:

- (i) neurons or channels;
- (ii) attention heads;
- (iii) MLP blocks;
- (iv) layer groups;
- (v) low-rank subspaces;
- (vi) structured matrix blocks;
- (vii) distributed cross-layer subspaces.

A key point is that a knowledge region need not be localized to one contiguous tensor block. Instead, define

$$P_K = \text{spanop}\{v_1, \dots, v_r\},$$

where the basis vectors may be distributed over multiple layers.

4 LLM Parameters as a Dynamical System

4.1 Discrete Dynamics

Continual learning naturally produces a discrete sequence:

$$\theta_{t+1} = \theta_t + \Delta\theta_t.$$

For localized plasticity,

$$\boxed{\theta_{t+1} = \theta_t + M_t \odot \Delta\theta_t.}$$

This is consistent with the RL-KPT formulation.

The update may be generated by gradient descent, an optimizer, a model-editing procedure, an RL controller, or a hybrid mechanism. The present theory does not require RL at this stage.

4.2 Continuous-Time Approximation

When updates are sufficiently frequent, introduce a continuous parameter time t :

$$\boxed{\frac{d\theta}{dt} = F(\theta, x, r, u, t)}$$

where:

- x is the incoming knowledge-bearing experience;
- r is feedback or reward;
- u is an optional control variable;
- F is the parameter evolution field.

For a fixed learning protocol and data stream, this simplifies to

$$\dot{\theta} = F(\theta, t).$$

This equation is the mathematical foundation of the proposed parameter dynamical system.

4.3 Local Linearization and Jacobian

Let θ^* be a local state. Define

$$J_\theta = \left. \frac{\partial F}{\partial \theta} \right|_{\theta^*}.$$

For a small perturbation $\delta\theta$,

$$\dot{\delta\theta} \approx J_\theta \delta\theta.$$

Hence

$$\delta\theta(t) \approx e^{J_\theta t} \delta\theta(0).$$

This equation gives a direct mathematical interpretation of the Jacobian: it determines the first-order local propagation of parameter-state perturbations.

5 Spectral State of the Parameter System

5.1 Parameter-Matrix Spectrum

For each parameter matrix W_l , define its singular-value decomposition:

$$W_l = U_l \Sigma_l V_l^\top,$$

where

$$\Sigma_l = \text{diag}(\sigma_{l,1}, \dots, \sigma_{l,r_l}), \quad \sigma_{l,1} \geq \dots \geq \sigma_{l,r_l} \geq 0.$$

For non-symmetric matrices, singular values are often more robust observables than eigenvalues. When W_l is symmetric or can be symmetrized appropriately, eigenvalues

$$\lambda_{l,1}, \dots, \lambda_{l,d_l}$$

may also be used.

The layer-level spectral descriptor can be written as

$$s_l = \left[\sigma_{l,1}, \dots, \sigma_{l,r_l}, \mu_l^{(1)}, \dots, \mu_l^{(p)} \right],$$

where $\mu_l^{(j)}$ denotes selected spectral moments.

The complete spectral state is

$$S_W = \Psi_W(W_1, \dots, W_L).$$

5.2 Jacobian Spectrum

For the parameter dynamics,

$$J_\theta = \frac{\partial F}{\partial \theta}.$$

Let its eigenvalues be

$$\lambda_1(J_\theta), \dots, \lambda_N(J_\theta).$$

The spectral radius is

$$\rho(J_\theta) = \max_i |\lambda_i(J_\theta)|.$$

For continuous-time local stability, the real parts of eigenvalues are central:

$$\max_i \operatorname{Re}(\lambda_i(J_\theta)).$$

For discrete dynamics with local transition matrix A , stability is associated with

$$\rho(A) < 1.$$

The distinction between parameter-matrix spectra and Jacobian spectra is essential:

Object	Mathematical role	Interpretation
W_l spectrum	Structure of a parameter matrix	Parameter subspace structure
J_θ spectrum	Local dynamics of parameter evolution	Stability, sensitivity, amplification
Hessian spectrum	Curvature of loss	Optimization geometry
Fisher spectrum	Information geometry	Parameter sensitivity to data distribution
Gradient covariance spectrum	Update directions	Dominant learning directions

The theory therefore does not reduce “spectral analysis” to one matrix. It defines a multi-spectral state.

6 Knowledge–Parameter–Spectral Mapping

6.1 Three Coupled Spaces

Define

$$\mathcal{K} \xleftarrow{\Phi} \mathcal{T} \xrightarrow{\Psi} \mathcal{S}.$$

That is,

$$K = \Phi(\theta), \quad S = \Psi(\theta, J_\theta).$$

The central hypothesis is that these spaces are coupled by the parameter state:

$$\boxed{K(t) \longleftrightarrow \theta(t) \longleftrightarrow S(t).}$$

This does *not* imply that Ψ is invertible. Spectral representations are generally many-to-one. Therefore, the framework distinguishes three different claims:

1. **Existence:** spectral observables contain information correlated with knowledge organization.
2. **Identifiability:** the relevant knowledge classes can be distinguished from the chosen spectral observables.
3. **Stability:** the correspondence remains sufficiently consistent under perturbations and across time.

Only the first is structurally expected from the parameter-encoding hypothesis. The second and third are empirical questions.

6.2 Dynamic Mapping

Let

$$K(t) = \Phi(\theta(t)), \quad S(t) = \Psi(\theta(t), J_\theta(t)).$$

Then the chain rule gives

$$\dot{K}(t) = D\Phi(\theta(t)) \dot{\theta}(t)$$

and

$$\dot{S}(t) = D\Psi \begin{bmatrix} \dot{\theta}(t) \\ \dot{J}_\theta(t) \end{bmatrix}.$$

Because

$$J_\theta(t) = D_\theta F(\theta(t), t),$$

its time derivative depends on higher-order derivatives of the dynamical field:

$$\dot{J}_\theta = D_\theta^2 F[\dot{\theta}] + D_{\theta t} F.$$

Thus knowledge evolution can, in principle, be studied through the induced motion of the spectral state.

6.3 A Useful Local Relation

Suppose a local spectral coordinate $s = \psi(\theta)$ is differentiable and the relevant Jacobian has full rank on a subspace $\mathcal{U}$. Then locally,

$$\delta s \approx D\psi(\theta) \delta \theta.$$

If $D\psi$ is injective on $\mathcal{U}$, parameter changes inside $\mathcal{U}$ are locally observable through s .

This does not prove that the spectral coordinate uniquely identifies knowledge. It provides a mathematical condition under which parameter dynamics can be detected through spectral observables.

7 Knowledge Regions as Spectral Subspaces

7.1 Distributed Knowledge Regions

For knowledge class K_i , define a parameter response operator

$$R_i(\theta) = \mathbb{E}_{x \sim K_i} [g(\theta, x)],$$

where g may be an activation response, gradient, sensitivity, or update response.

The corresponding dominant subspace can be obtained from

$$C_i = \mathbb{E}_{x \sim K_i} [g(\theta, x)g(\theta, x)^\top].$$

Let

$$C_i = Q_i \Lambda_i Q_i^\top.$$

The leading eigenvectors of C_i define a knowledge-responsive subspace

$$\mathcal{U}_i = \text{span}\{q_{i,1}, \dots, q_{i,r_i}\}.$$

This provides an alternative to the stronger and unnecessary claim that knowledge corresponds to a contiguous set of individual parameters.

7.2 Knowledge Similarity and Subspace Similarity

For two knowledge domains K_i, K_j , define the principal angles

$$0 \leq \alpha_1 \leq \dots \leq \alpha_r \leq \frac{\pi}{2}$$

between $\mathcal{U}_i$ and $\mathcal{U}_j$.

Small principal angles indicate strong overlap:

$$\mathcal{U}_i \approx \mathcal{U}_j.$$

Large angles indicate relative separation.

A subspace similarity measure may be defined as

$$\text{Sim}(i, j) = \frac{1}{r} \sum_{k=1}^r \cos^2 \alpha_k.$$

This yields a mathematical mechanism for studying interdisciplinary knowledge:

$$K_{\text{robotics}} \sim K_{\text{mechanics}}, K_{\text{control}}, K_{\text{physics}}, K_{\text{mathematics}}.$$

8 Spectral Perturbation and Knowledge Change

Suppose a new knowledge item induces

$$W \rightarrow W + \Delta W.$$

For a symmetric matrix, Weyl-type perturbation bounds give

$$|\lambda_i(W + \Delta W) - \lambda_i(W)| \leq \|\Delta W\|_2.$$

For singular values,

$$|\sigma_i(W + \Delta W) - \sigma_i(W)| \leq \|\Delta W\|_2.$$

Therefore, if knowledge acquisition causes a nonzero parameter change, its spectral consequences can be bounded.

This provides a principled reason to measure spectral changes rather than simply plotting raw parameter differences.

Define the spectral displacement

$$D_S(W, W') = \|\Sigma(W) - \Sigma(W')\|_2,$$

or, for a collection of layers,

$$D_S^{(L)} = \sum_{l=1}^L w_l \|\Sigma_l(W'_l) - \Sigma_l(W_l)\|_2.$$

Similarly, define a parameter displacement

$$D_\theta = \|\theta' - \theta\|_2.$$

A knowledge-sensitive spectral response can then be tested through

$$\Delta K \longrightarrow \Delta \theta \longrightarrow \Delta S.$$

9 Dynamical Stability and Knowledge Stability

The word “stability” requires careful interpretation.

A knowledge region need not be permanently stable. In continual learning, the knowledge topology itself may evolve:

$$P_i(t+1) \neq P_i(t).$$

The appropriate question is therefore not whether

$$P_i(t) = P_i(0)$$

for all t , but whether its evolution is controlled.

Define a knowledge-region drift:

$$D_P(i; t, t + \Delta t) = d(\mathcal{U}_i(t), \mathcal{U}_i(t + \Delta t)),$$

where d may be based on principal angles.

Define spectral drift:

$$D_S(t, t + \Delta t) = d_S(S(t), S(t + \Delta t)).$$

A continuously learning system may be regarded as locally stable if:

$$D_P(i; t, t + \Delta t) \leq \epsilon_i$$

for knowledge classes that should remain preserved, while allowing larger changes for newly introduced or intentionally reorganized knowledge.

This leads to a more useful notion:

controlled plasticity rather than absolute stability.

10 Other Mathematical Observables

10.1 Hessian Geometry

Let the training loss be $L(\theta)$ and define

$$H(\theta) = \nabla_{\theta}^2 L(\theta).$$

The Hessian spectrum describes local curvature of the optimization landscape.

Knowledge-specific curvature can be estimated from

$$H_i = \mathbb{E}_{x \sim K_i} [\nabla_{\theta}^2 \ell(\theta; x)].$$

Comparing H_i and H_j provides another route to distinguish knowledge-related parameter directions.

10.2 Fisher Information

For model distribution $p_{\theta}(y|x)$, define

$$F(\theta) = \mathbb{E} \left[\nabla_{\theta} \log p_{\theta}(y|x) \nabla_{\theta} \log p_{\theta}(y|x)^{\top} \right].$$

The Fisher information matrix identifies parameter directions that are sensitive to changes in the model's predictive distribution.

A knowledge-conditioned Fisher matrix

$$F_i = \mathbb{E}_{x \sim K_i} \left[\nabla_{\theta} \log p_{\theta} \nabla_{\theta} \log p_{\theta}^{\top} \right]$$

can therefore be used to identify knowledge-sensitive directions.

10.3 Gradient-Covariance Geometry

For each knowledge class,

$$G_i = \mathbb{E}_{x \sim K_i} [g_x g_x^\top], \quad g_x = \nabla_\theta \ell(\theta; x).$$

The dominant eigenspace of G_i describes the directions in parameter space most strongly involved in learning that knowledge class.

This gives a direct bridge between the RL-KPT notion of “where to update” and a mathematically measurable subspace:

$$P_i \approx \text{TopEig}(G_i).$$

10.4 Information-Theoretic View

Let Z_S denote a spectral representation and Z_K a knowledge representation. The mutual information

$$I(Z_S; Z_K)$$

can be used as an empirical measure of how much knowledge-class information is contained in spectral observables.

A successful knowledge-spectral representation should satisfy

$$I(Z_S; Z_K) > 0,$$

while useful classification requires stronger conditions such as separability and generalization across held-out knowledge instances.

11 A Unified Knowledge–Parameter Dynamical Model

The preceding concepts can be summarized by the state

$$\mathcal{X}(t) = (\theta(t), K(t), S(t), J_\theta(t)).$$

The coupled system is

$$\begin{aligned} \dot{\theta} &= F(\theta, x, r, u), \\ K &= \Phi(\theta, x), \\ J_\theta &= D_\theta F, \\ S &= \Psi(\theta, J_\theta), \\ \dot{K} &= D\Phi \dot{\theta}, \\ \dot{S} &= D\Psi(\dot{\theta}, \dot{J}_\theta). \end{aligned}$$

This is the proposed *LLM Parameter Dynamical System*.

It contains four conceptually distinct layers:

1. **State layer:** parameter state θ ;